

Evaluation of Radiation Exposure in Pediatric Interventional Cardiology Procedures

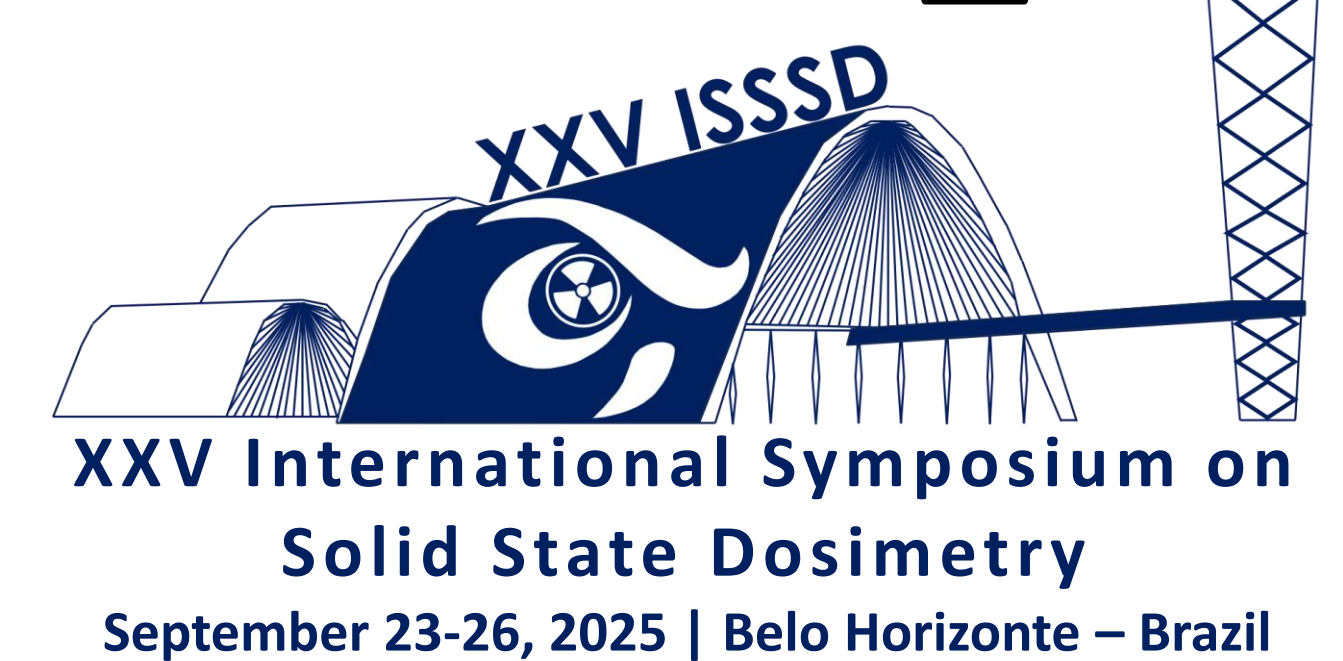
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1. INTRODUCTION

➤ The establishment of diagnostic reference levels (DRLs) is essential in radiology, particularly in pediatric interventional cardiology. These levels serve as a benchmark for assessing radiological practices, enabling the identification of procedures that result in radiation doses exceeding recommended limits. In interventional cardiology, the main quantities used to define DRLs are the kerma-area product (P_{KA}) and the air kerma at the reference point (K_{a,r}), with fluoroscopy time also commonly considered. In pediatrics, it is recommended that DRLs be evaluated according to weight ranges, as significant weight variation can exist within the same age group. Interventional cardiology procedures can be classified as diagnostic or therapeutic, depending on their purpose. This study aimed to determine the dosimetric quantities used to establish DRLs in a pediatric interventional cardiology service.

2. MATERIALS AND METHODS

➤ Data were collected from January 2019 to April 2025 from dose reports generated by the imaging equipment at the end of each procedure, and were categorized by patients' weight and age groups. The values were corrected using a previously calculated correction factor².

3. RESULTS AND DISCUSSION

➤ The data were statistically processed using Origin software. Figure 1 presents the P_{KA} values obtained as a function of patients' age (a) and body weight (b).

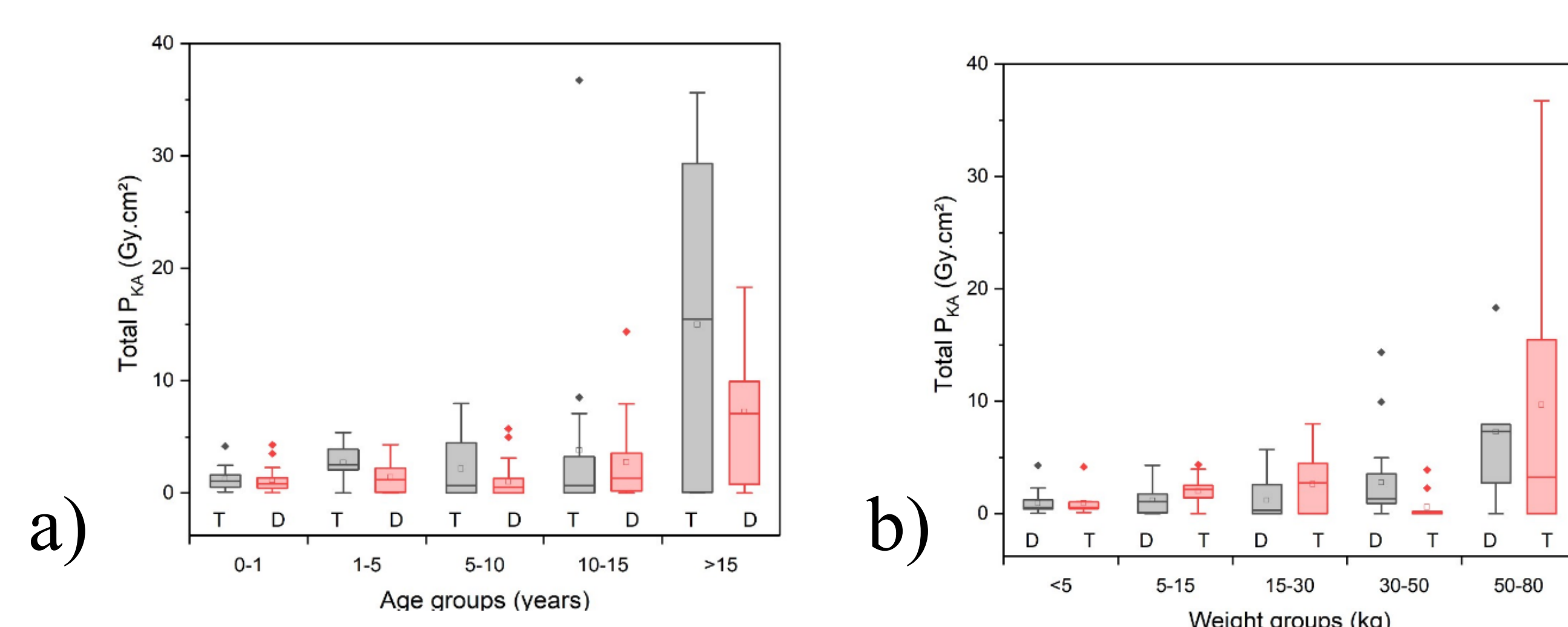


Figure 1: P_{KA} according to the age (a) and weight (b) of the evaluated patients.

- It was observed that therapeutic procedures generally resulted in higher radiation doses across most evaluated groups when compared to diagnostic procedures. The dose values obtained were also consistent with those reported in previous studies.
- The evaluation of total P_{KA} (Gy·cm²) by age groups showed significant dose variation across diagnostic and therapeutic procedures. Higher P_{KA} values were mainly observed in patients >15 years, likely reflecting more complex procedures and larger body size, while younger groups (0–1 and 1–5 years) exhibited lower and more uniform values.

4. CONCLUSIONS

- This study analyzed radiation dose levels in 215 pediatric interventional cardiology procedures. Significant differences in dose-area product (P_{KA}) were found between diagnostic and therapeutic procedures, with higher doses in therapeutic cases. Dose values increased with patient age and weight, reflecting greater technical demands and procedure duration.

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REFERENCES

- [1] ICRP, 2017. Diagnostic Reference Levels in Medical Imaging. ICRP Publication 135. Annals of the ICRP, 46(1). <https://doi.org/10.1177/0146645317717209>.
- [2] Yagui, A., Schelin, H., Mello, R., Andrade, D., Vosiak, P., Denyak, V., Filipov, D., Paschuk, S., Khoury, H., & Ubeda, C. (2024). Establishment of local diagnostic reference levels for radiation dose in paediatric interventional cardiology procedures in Brazil. Radiation Physics and Chemistry. <https://doi.org/10.1016/j.radphyschem.2024.111577>